



## CONTACTS

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## LINKS

t.me/IgorsVolohovs

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github.com/IGORSVOLOHOVS

## SKILLS

- C++23Python 3.13
- Qt FrameworkWindows API
- Visual StudioCMake
- Github & ActionsSWIG
- Google AntigravityLinux
- ConanFreeRTOS
- UML 2.0 / C4 Model
- Jira / Confluence
- Rust / Java / C# / Julia
- OpenGLSTM32CubeIDE
- GTest / GBenchmark
- Docker / n8nJenkinsSQL

## LANGUAGES

|         |              |
|---------|--------------|
| English | Professional |
| Latvian | Native       |
| Russian | Native       |

## HOBBIES

- Assemble a 3×3 - 7×7 Rubik's Cube
- Learning Maths & Developing my own "Theory of effective thinking and task solving"

# IGORS VOLOHOVS

C/C++ Developer

## ABOUT ME

C++ Developer focused on solving complex technical challenges and low-level system optimization. I am a proactive learner who enjoys deep-diving into technology and sharing knowledge with the team.

## WORK EXPERIENCE

**Software Developer in TSI AdditiveLab**

Jan 2024 - Present

TSI | Riga, Latvia (Wire Arc Additive Manufacturing)

Developing **WaamMonitoring** — a comprehensive software for real-time monitoring, graphical analysis, and data storage of the metal 3D printing process (my Bachelor's thesis project). Implemented **WaamAPI**: a low-level HAL in C++23 with Python 3.13 bindings for seamless hardware-software integration.

**Hardware:** Yaskawa GA-50, Yaskawa HC-10, Wenglor MLZL171 (2D profilometer), Optris CTi Pyrometers, Fronius TPS 500i.

**Technologies used:** C++23, Python 3.13, Qt Framework, Windows API.

**Software Developer (Edge AI)**

Jan 2026 - Present

TSI AdditiveLab | Riga, Latvia

Edge AI device based on Rockchip RV1106. Preparation of dataset, training and deployment of YOLOv8n neural network for defects localization. The goal of this project is to create a low-cost Edge AI device (robot mounted) to control quality of welding seems in AdditiveLab.

**Technologies used:** YOLOv8n, Python, C++, OpenCV, SSH, RKNN-Toolkit2.

**Medical Device Embedded Firmware Developer**

Dec 2022 - Present

IDEAHUB | Riga, Latvia

Development of firmware for a medical cell staining machine. Connected various sensors, motors and network devices to STM32F4 MCUs, and wrote a driver for UART-USB module to connect Raspberry Pi with several STM32F4 MCUs.

**Technologies used:** C/C++11, STM32CubeIDE, Linux, Rust, SWIG.

**ESF Project Developer**

Feb 2023 - Jul 2023

TSI | Riga, Latvia

Creating software for the mobile robot KOALA. The goal was to create a user-friendly development environment in C/C++ and Python for creating new projects.

**Technologies used:** CMake, C/C++11, Python, Linux.

**Private Tutor (Math & Programming)**

Sep 2022 - Present

Teaching Mathematics and Programming to schoolchildren, helping them build strong logical foundations and practical coding skills.

## EDUCATION

○

**Computer Vision Course**

Present

Yandex Practicum

2 projects completed; working with OpenMMLab, PyTorch, ClearML, CVAT.

○

**C++ Middle Developer**

2025

Yandex Practicum

9 projects; advanced C++20/23 and introduction to C++26 standards.

○

**Computer Science Bachelor Program**

2021 - Present

TSI, Riga

○

**C++ Developer (~300 tasks were solved)**

2021

Yandex Practicum

○

**Basic Course for Network Management**

2019

ITLAT, Riga

○

**Basic Course of Java**

2018

ITLAT, Riga

○

**Basic Course of C/C++**

2017

ITLAT, Riga